

The Optimal Quantity of Capital and Debt

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Introduction

Revisit classic questions in public economics:

- What is the optimal amount of capital and government debt?
- Should capital be taxed? If yes, how much?
- What is the optimal amount of redistribution?

Incomplete Market: Aiyagari (1995)

- Theoretical findings
- New quantitative method

Our Approach

Use Aiyagari (1995) model and framework

- Heterogenous agents incomplete markets model.
- Capital, government debt and endogenous labor supply.
- Ramsey taxation approach.
- Policy Instruments: linear labor tax, linear capital tax, government debt (and transfer).
- Full commitment.
- Important: Maximize period 0 welfare.

Theoretical Contribution

1. The optimal **capital** satisfies the modified golden rule: return on capital equals time discount rate.
2. The long-run steady state level of **government debt** is independent of initial condition (such as initial debt).

Methodological Contribution

- Objective:

Compute optimal policy along the full transition path.

- Independence from initial conditions enables to do this.

1. Solve for the optimal steady-state — **terminal condition**.

Initial condition: current US economy and policy.

2. Solve for the **transition path** using the initial and terminal conditions.

Quantitative Contribution

- Compute optimal policy along the full transition path.
- We find (robust across various calibrations):
 1. Government debt: Increases over time; High long-run level.
 2. Capital tax: High initially; Low in the long run.
 3. Labor tax: Low initially; High in the long run.

Literature

- Aiyagari (1995)
 - Ramsey problem, Lagrangian approach
 - Capital: modified golden rule
- Aiyagari and McGrattan (1998)
 - Maximize steady-state welfare
 - Debt: liquidity
- Heathcote and Tsujiyama (2021)
 - One-time tax change
 - Transfers and tax regressivity

Literature

- Le Grand and Ragot (2020)
 - Lagrangian approach, derive FOCs
 - Perturbation around the initial SS
 - Different models: optimal UI, firm taxation
- Dyrda and Pedroni (2021)
 - Directly search for tax sequences maximizing welfare
 - Our approach can be used to find optimal SS first

The Bewley-Imrohoroglu-Huggett-Aiyagari Model

- Time: $t \in \{0, 1, 2, \dots\}$
- Households
 - A continuum of households
 - Idiosyncratic labor productivity shocks $e_t \in E$, following a Markov process
 - Access to an incomplete market: risk-free asset $a_t \in A$
 - Borrowing constraint: $a_t \geq -\underline{a}$
- Standard representative firm
- Government: Ramsey planner with full commitment

Government

- Linear capital income tax rate τ_{kt} .
- Linear labor income tax τ_{nt} .
- Issue debt B_t .
- Lump-sum transfer T_t (not a choice in benchmark).
- Finance exogenous government spending G_t .

Markets

- State of economy is μ_t , a measure on $S = E \times A$.
- Aggregation:

$$A_t = \int_S a_t d\mu_t,$$

$$N_t = \int_S e_t n_t d\mu_t,$$

$$K_t = A_t - B_t.$$

- Prices

$$r_t = F_K(K_t, N_t) - \delta,$$

$$w_t = F_N(K_t, N_t),$$

- After-tax prices

$$\bar{r}_t = (1 - \tau_{kt}) r_t,$$

$$\bar{w}_t = (1 - \tau_{nt}) w_t.$$

Household's Problem

$$V_0(a_0, e_0) = \max_{\{a_{t+1}, c_t, n_t\}} \mathbb{E}_0 \sum_{t=0}^{\infty} \beta^t u(c_t, n_t)$$

subject to

$$\begin{aligned} c_t + a_{t+1} &\leq a_t(1 + \bar{r}_t) + \bar{w}_t e_t n_t + T_t, \\ a_{t+1} &\geq -\underline{a}, \end{aligned}$$

where

$$u(c, n) = \frac{\left(c - \chi \frac{n^{1+\frac{1}{\phi}}}{1+\frac{1}{\phi}}\right)^{1-\sigma}}{1-\sigma}$$

Ramsey Problem

$$\max_{\{\bar{r}_t, \bar{w}_t, K_t, B_{t+1}, a_{t+1}, c_t\}} \int V_0(a_0, e_0) d\mu_0$$

subject to $\{\bar{r}_t, \bar{w}_t, K_t, B_{t+1}, a_{t+1}, c_t\}$ be a competitive equilibrium

1. Households maximize utility (take prices and policies as given).
2. Firms maximize profits (take prices and policies as given).
3. Markets clear. Resource constraint is satisfied.

Important: Ramsey planner constraints include

- Consumption Euler Equation:

$$u_c(c_t, n_t) \geq \beta(1 + \bar{r}_{t+1}) \mathbb{E}_t u_c(c_{t+1}, n_{t+1}).$$

- Household credit constraint

$$(a_{t+1} + \underline{a})(u_c(c_t, n_t) - \beta(1 + \bar{r}_{t+1}) \mathbb{E}_t u_c(c_{t+1}, n_{t+1})) = 0.$$

Lagrangian

$$\begin{aligned}\mathcal{L} = & \int \mathbb{E}_0 \sum_{t=0}^{\infty} \beta^t \left\{ u(c_t, n_t) \right. \\ & + \theta_{t+1} \left(u_c(c_t, n_t) - \beta(1 + \bar{r}_{t+1}) E_t u_c(c_{t+1}, n_{t+1}) \right) \\ & + \eta_{t+1} (a_{t+1} + \underline{a}) \left(u_c(c_t, n_t) - \beta(1 + \bar{r}_{t+1}) E_t u_c(c_{t+1}, n_{t+1}) \right) \\ & \left. + \gamma_t \left(F(K_t, N_t) - \delta K_t + B_{t+1} - G_t - T_t - (1 + \bar{r}_t) B_t - \bar{r}_t K_t - \bar{w}_t N_t \right) \right\} d\mu_0\end{aligned}$$

↪ Yields sequences of FOCs for all variables

$$\{\bar{r}_t, \bar{w}_t, K_{t+1}, B_{t+1}, a_{t+1}, c_t\}.$$

Details

FOC Example

$$\partial \mathcal{L} / \partial B_{t+1} :$$

$$\gamma_t = \beta (F_K(K_{t+1}, N_{t+1}) + 1 - \delta) \gamma_{t+1}.$$

$$\vdots$$
$$\vdots$$
$$\vdots$$
[Details](#)

Assumption

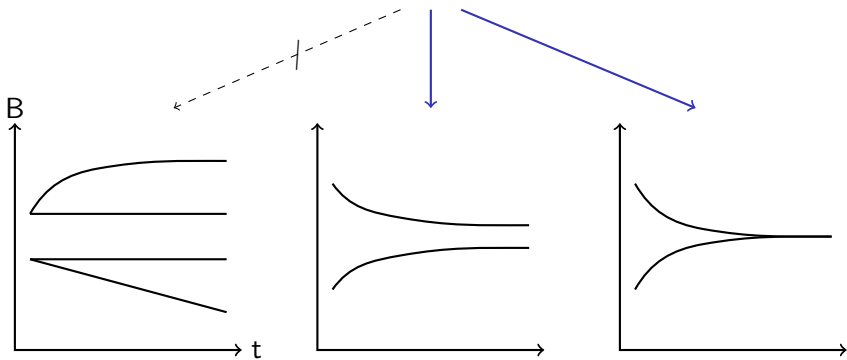
Assumption

For each set of initial conditions (B_0, K_0, μ_0) , the economy (including policy and all other variables) converges to a unique steady state.

Remarks:

- Convergence includes policies and other variables such as Lagrange multipliers.
- Key: convergence to steady-state.
- Uniqueness assumed given a set of initial conditions (B_0, K_0, μ_0) .
- Uniqueness arises generically in maximization problems.

Assume



Prove

Theoretical Result I: Steady State Capital

Theorem

Steady state capital satisfies the modified golden rule:

$$\beta (F_K(K, N) + 1 - \delta) = 1$$

- FOC:

$$\gamma_t = \beta (F_K(K_{t+1}, N_{t+1}) + 1 - \delta) \gamma_{t+1}$$

- Efficient allocations of resources across time.
- Government debt needed
 - ↪ absorbs additional precautionary savings.
- Why not capital $\uparrow \Rightarrow$ wages $\uparrow \Rightarrow$ consumption poor $\uparrow$?
 - ↪ capital tax $\downarrow \Rightarrow$ labor tax $\uparrow \Rightarrow$ bad for labor earners.

Theoretical Result II: Independence of Initial Conditions

Theorem

*The long-run values of **government debt**, of the labor income tax rate and of the capital income tax rate are generically independent of the initial level of government debt (and the initial capital stock).*

Intuition: Independence

- Complete Markets:

- Not enough independent optimality conditions to determine the long-run steady-state.
- Bonds have no real value.
- One equation (government budget constraint) to determine both bonds B and labor tax τ_n

- Incomplete Markets:

- Bonds have real value.
- Provides additional equation.

Equations

Economic Intuition: Role of Debt

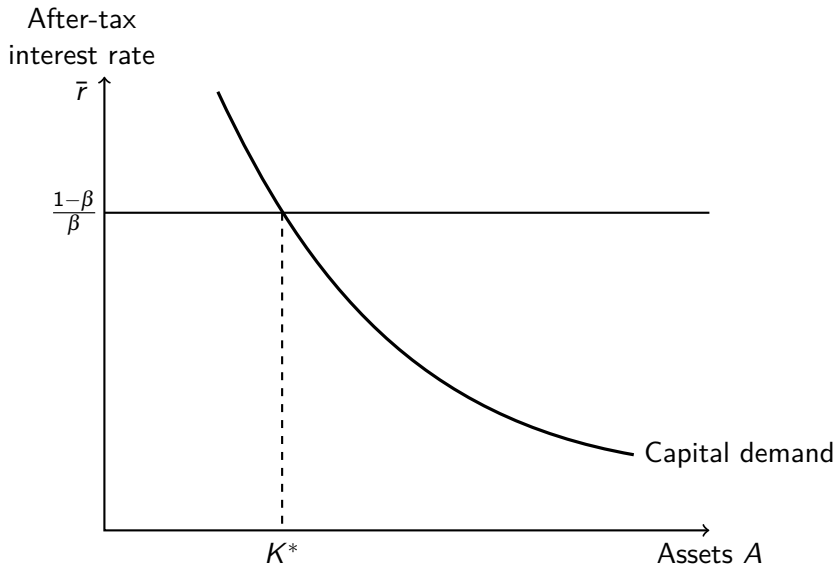
- Complete Markets:

- Only one role for debt: smooth tax distortions.
- Benefit: lower initial distortions; Cost: higher distortions later.
 - ↪ Labor tax and debt are (basically) constant over time.
 - ↪ Steady-state debt depends on initial level of debt.

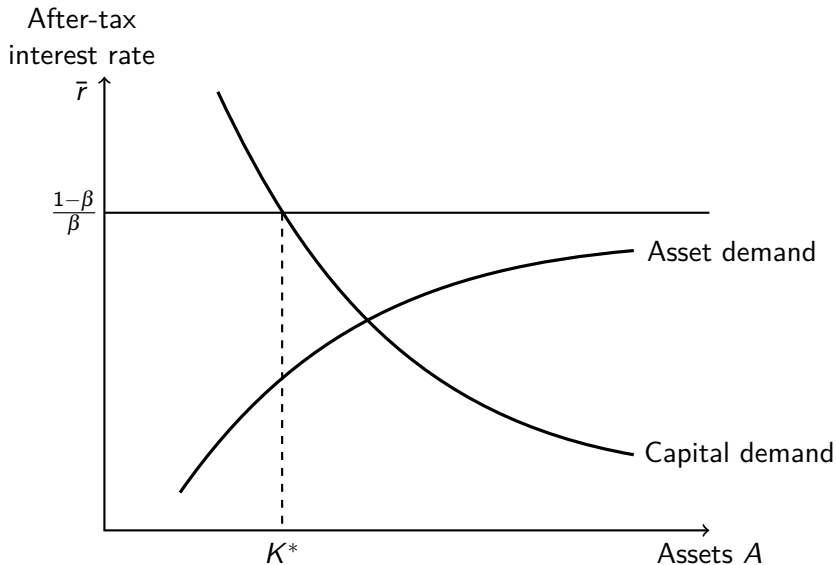
- Incomplete Markets:

- Additional role: smooth consumption (relax borrowing constraint).
 - ↪ Independent of initial level of debt.
 - ↪ Additional benefit: more debt, higher liquidity
- Optimum level of debt: marginal benefits = marginal costs.

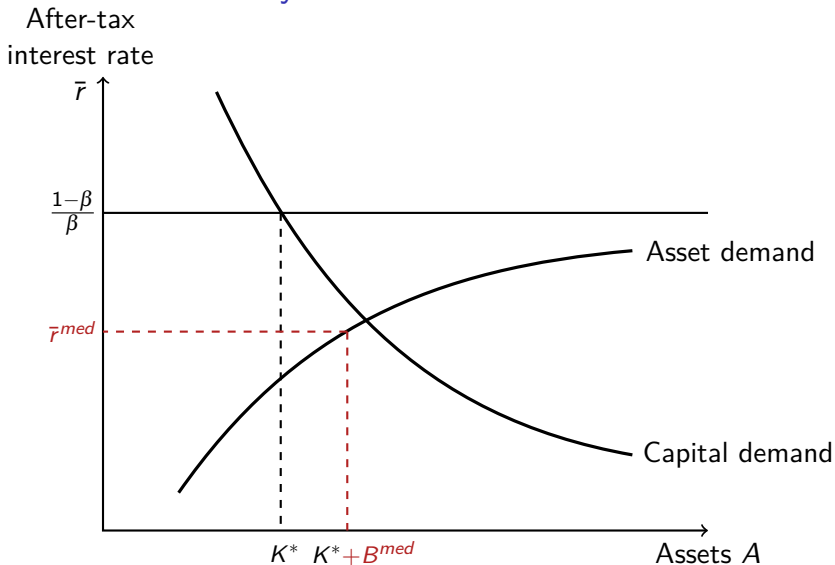
The Additional Equation



The Additional Equation

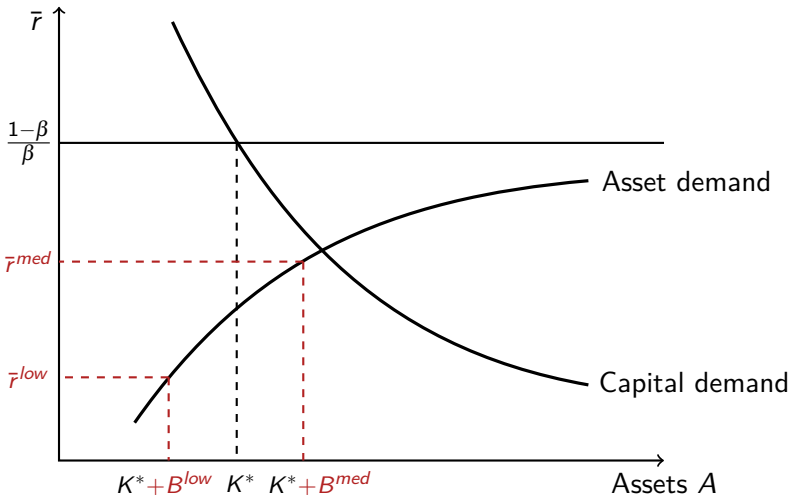


Ramsey Planner Trade-off



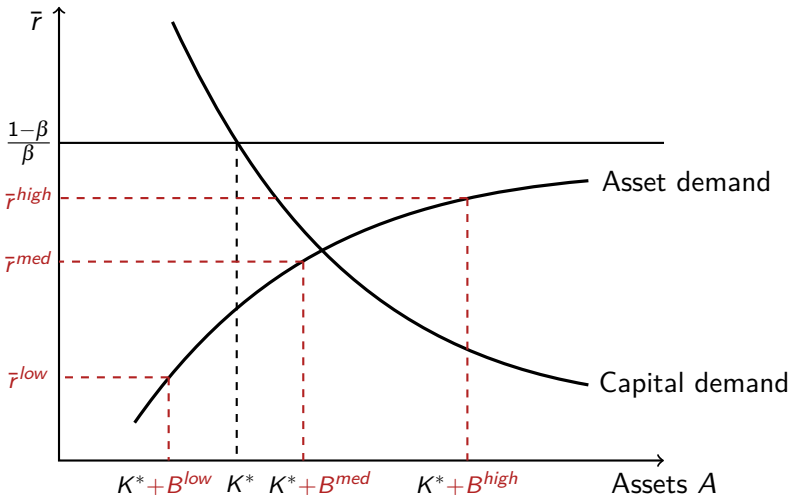
Ramsey Planner Trade-off

After-tax
interest rate

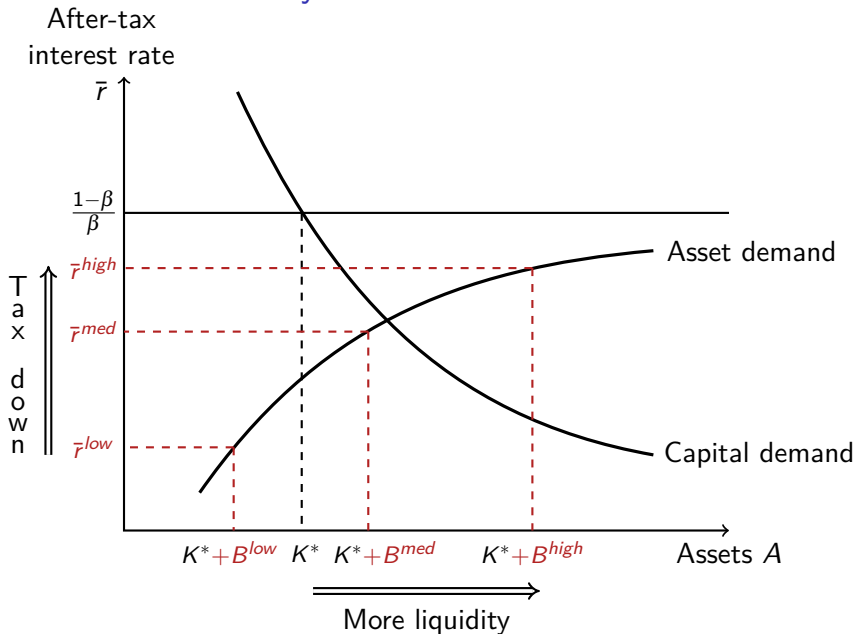


Ramsey Planner Trade-off

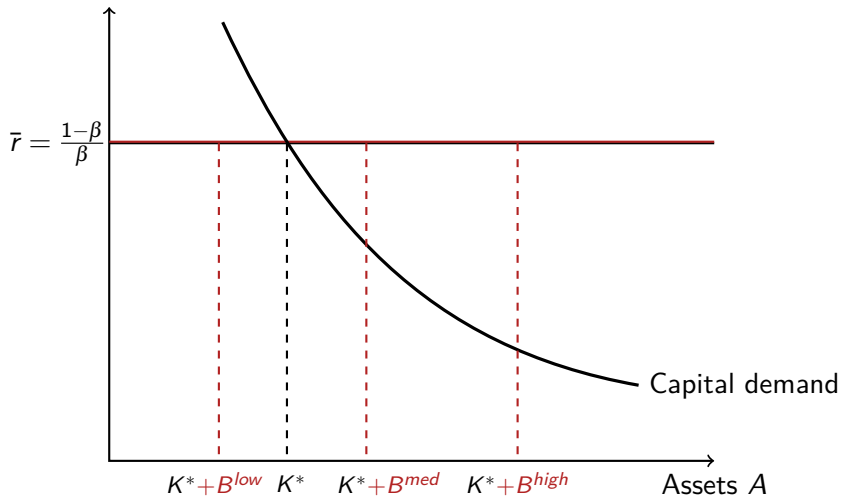
After-tax
interest rate



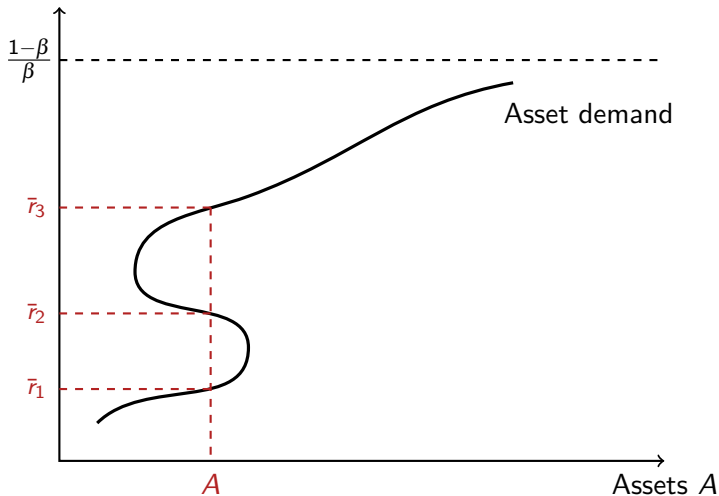
Ramsey Planner Trade-off



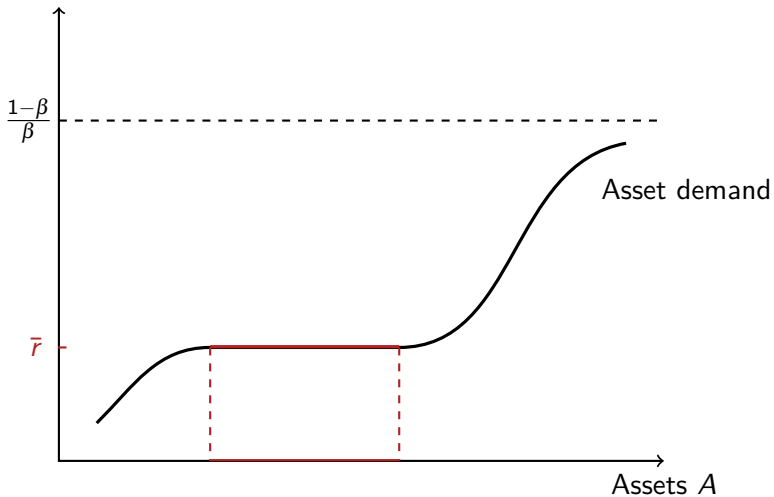
Complete Markets: No Such Trade-Off



Finite Number of Solutions



Non-Generic Indeterminacy



Sketch of the Formal Proof

Step 1: Formalize the intuition:

Finite number of solutions to the steady state FOCs
(key deviation from the representative agent model)

Step 2: Apply Berge's Maximum Theorem:

The optimal policy path (including the steady state policy) is continuous in the initial condition
(government debt)

Step 3: A discrete continuous function is constant:

The optimal steady state is independent of
the initial condition.

Quantitative Exercise

- Apply the quantitative methodology to Aiyagari model calibrated to the U.S. economy.
- Solve for the optimal steady state. Algorithm
- Solve for the optimal transition path.
(Using the independence result). Algorithm

Quantitative Analysis: Calibration

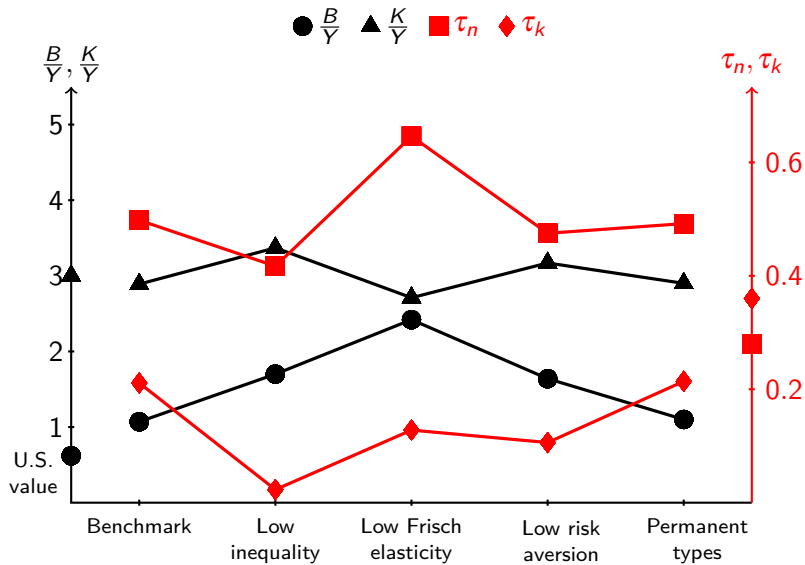
Para.	Value	Description	Source/Target
Exogenous Parameters			
σ	2(1)	Coeff. of risk aversion	Trabandt and Uhlig (2011)
ϕ	1(0.5)	Frisch elasticity	
α	0.36	Capital share	
δ	0.08	Depreciation rate	
τ_l	28%	Labor inc. tax rate	Holter et al. (2015)
τ_k	36%	Capital inc. tax rate	
$\frac{B}{Y}$	62%	Debt to GDP ratio	Prescott (2004)
$\frac{G}{Y}$	7.3%	Gov. exp. to GDP ratio	
Calibrated Parameters			
ρ	0.94(0.6)	Persist. of labor prod.	$a_{90}/a_{50} = 7.55$
σ_u	0.18(0.2)	S.D. of labor prod. shock	$var(\log y) = 1.3$
β	0.96	Discount rate	$K/Y = 3$
χ	2.56	Disutility from labor	$mean(n) = 0.33$

Steady State Results

	Optimal Steady State (1)	U.S. Calibration (2)
τ_l	49.8%	28.0%
τ_k	21.1%	36.0%
T/Y	21.1%	14.1%
B/Y	1.07	0.62
K/Y	2.89	3.00
$\int n$	0.23	0.33
$N = \int en$	0.31	0.46
$var(\log(y + T))$	0.30	0.51
$var(\log(y + T + \bar{r}a))$	0.35	0.52
coeff. var. a	1.37	1.52
coeff. var. c	0.64	0.80

[Details](#)

Robustness



Robust Conclusions

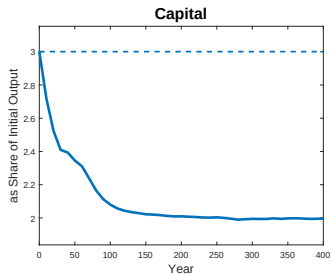
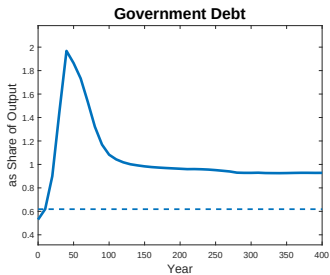
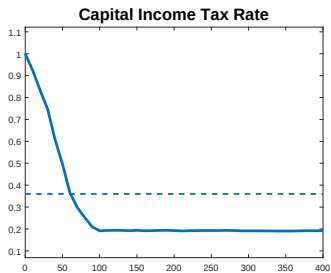
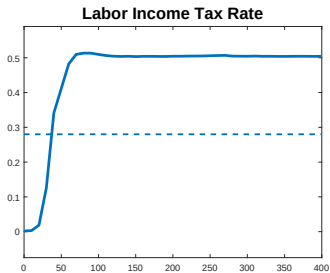
- High level of government debt.
 - ↪ Tax distortions do not put tight bounds on the level of government debt.
- Low level of capital income taxes.
 - ↪ Large amount of debt requires an after tax interest rate close to the rate of time preference.
- High level of labor tax.
 - ↪ High interest rate payments.
- Reduce inequality:
 - More debt and low capital income tax rates.
 - Not: Low debt and high capital income tax rates.

Transition: Computational Strategy

1. Given after-tax prices, solve for the households' policy functions.
2. Simulate the economy forward, starting from the initial distribution of households, to compute assets and labor.
3. Compute capital and debt backward.
4. Solve for the Lagrange multipliers backward.
5. Simulate the economy forward to check the FOCs w.r.t. after-tax prices.
6. Update after-tax prices until convergence.

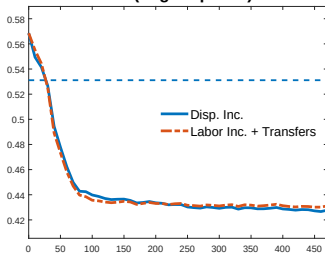
Each period is 10 years, because of slow convergence of distribution over time.

Transition Path

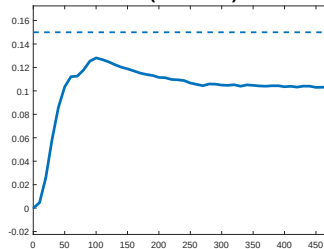


Inequalities along the Transition

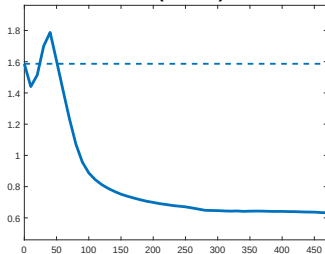
Var (Log Disp. Inc.)



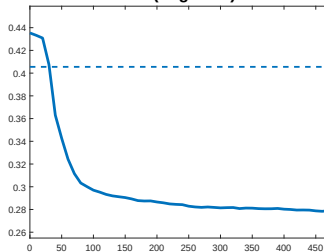
Var(Asset Inc.)



Var (Assets)



Var (Log Con.)



Welfare Analysis

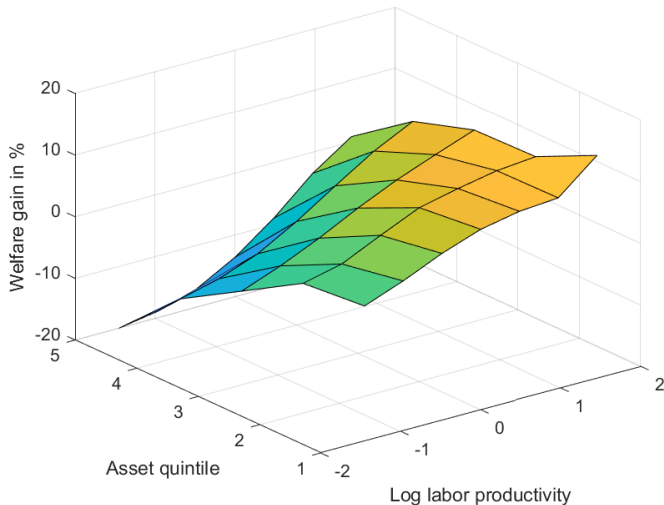
- Solve for the consumption equivalence gain φ :

$$V_0^{ce}(a_0, e_0) = \mathbb{E}_0 \sum_{t=0}^{\infty} \beta^t \frac{\left(\varphi c_t^{ini} - \chi \frac{(n_t^{ini})^{1+\frac{1}{\phi}}}{1+\frac{1}{\phi}} \right)^{1-\sigma}}{1-\sigma}$$

$$\int V_0^{ce}(a_0, e_0) d\mu_0 = \int V_0^{opt}(a_0, e_0) d\mu_0,$$

- $\varphi = 2.59\%$
- φ decreasing in wealth, increasing in labor productivity

Welfare Gain in Consumption Equivalents by Productivity Level and Wealth



Conclusion

- Theoretical Results:
 - Long-run capital stock satisfies modified golden rule.
 - Long-run level of government debt is **independent of initial conditions**.
- Methodological Results:
 - New computational method, using Lagrangian.
 - Can **first compute the long run steady states**.
 - Then compute the transition from the initial to the long-run steady state.
- (Robust) Quantitative Results:
 - Government debt: High in the long-run.
 - Capital tax: High initially; Low in the long-run.
 - Labor tax: Low initially; High in the long-run.

Additional Slides

Details on Household's Problem

- Euler equation

$$u_c(c_t, n_t) \geq \beta(1 + \bar{r}_{t+1}) \mathbb{E}_t[u_c(c_{t+1}, n_{t+1})]$$

- Kuhn-Tucker condition

$$(a_{t+1} + \underline{a})(u_c(c_t, n_t) - \beta(1 + \bar{r}_{t+1}) \mathbb{E}_t[u_c(c_{t+1}, n_{t+1})]) = 0$$

Details on the Lagrangian

- Let θ_{t+1} and $-\eta_{t+1}$ represent the Lagrange multipliers for HH EE and KT
- $\theta_{t+1} u_c(c_t, n_t), -(1 + \bar{r}_{t+1}) \theta_{t+1} u_c(c_{t+1}, n_{t+1})$
- $-\eta_{t+1}(a_{t+1} + \underline{a}) u_c(c_t, n_t), (1 + \bar{r}_{t+1}) \eta_{t+1}(a_{t+1} + \underline{a}) u_c(c_{t+1}, n_{t+1})$
- Then define $\lambda_{t+1} \equiv \eta_{t+1}(a_{t+1} + \underline{a}) - \theta_{t+1}$
- The Lagrangian is

$$\mathcal{L} = \int \mathbb{E}_0 \sum_{t=0}^{\infty} \beta^t \left\{ (u(c_t, n_t) + u_c(c_t, n_t)(\lambda_t(1 + \bar{r}_t) - \lambda_{t+1})) \right. \\ \left. + \gamma_t \left(F(K_t, N_t) - \delta K_t + B_{t+1} - G_t - T_t - (1 + \bar{r}_t) B_t - \bar{r}_t K_t - \bar{w}_t N_t \right) \right\} d\mu_0.$$

subject to budgets constraints, borrowing constraints, resource constraints, and market clearing conditions.

Interior Solution

The interior solution satisfies the following conditions

$$\lambda_{t+1} : u_{c,t} \geq \beta (1 + \bar{r}_{t+1}) \mathbb{E}[u_{c,t+1}]$$

with equality if $a_{t+1} > -\underline{a}$

$$\begin{aligned} a_{t+1} : & u_{c,t} - \frac{\partial c_t}{\partial a_{t+1}} u_{cc,t} [\lambda_t (1 + \bar{r}_t) - \lambda_{t+1}] \\ &= \beta \mathbb{E} \left[(1 + \bar{r}_{t+1}) u_{c,t+1} + \frac{\partial c_{t+1}}{\partial a_{t+1}} u_{cc,t+1} [\lambda_{t+1} (1 + \bar{r}_{t+1}) - \lambda_{t+2}] \right] \\ &+ \beta \gamma_{t+1} (F_K(K_{t+1}, N_{t+1}) - \delta - \bar{r}_{t+1}) \\ &\text{if } a_{t+1} > -\underline{a}, \text{ otherwise } \lambda_{t+1} = 0 \end{aligned}$$

$$B_{t+1} : \gamma_t = \beta (F_K(K_{t+1}, N_{t+1}) + 1 - \delta) \gamma_{t+1}$$

Interior Solution II

$$\begin{aligned}\bar{r}_t : \gamma_t A_t &= \gamma_t (F_{N,t} - \bar{w}_t) \frac{\partial N_t}{\partial \bar{r}_t} \\ &+ \mathbb{E}_t \left[u_{c,t} \lambda_t + a_t u_{c,t} + \frac{\partial c}{\partial \bar{r}} u_{cc,t} [\lambda_t (1 + \bar{r}_t) - \lambda_{t+1}] \right] \\ \bar{w}_t : \gamma_t N_t &= \gamma_t (F_{N,t} - \bar{w}_t) \frac{\partial N_t}{\partial \bar{w}_t} \\ &+ \mathbb{E}_t \left[e_t n_t u_{c,t} + \frac{\partial c}{\partial \bar{w}} u_{cc,t} [\lambda_t (1 + \bar{r}_t) - \lambda_{t+1}] \right]\end{aligned}$$

together with borrowing and budget constraints and market clearing conditions.

Back

Additional Equation in Incomplete Markets

- Complete Market:

$$\begin{aligned}\gamma &= \beta(1+r)\gamma, \\ u_c &= \beta(1+\bar{r})u_c, \\ \cancel{u_c + \frac{\partial c}{\partial a'} u_{cc} \lambda \bar{r}} &= \cancel{\beta(1+\bar{r}_{t+1}) \left(u_c + \frac{\partial c}{\partial a'} u_{cc} \lambda \bar{r} \right)} + \beta\gamma(r - \bar{r}).\end{aligned}$$

- Incomplete Markets:

$$\begin{aligned}\gamma &= \beta(1+r)\gamma, \\ u_c &\geq \beta(1+\bar{r})\mathbb{E}[u'_c], \\ -u_{cc}[\lambda(1+\bar{r}_t) - \lambda'] &= \beta\mathbb{E}[-(1+\bar{r})u'_{cc}[\lambda'(1+\bar{r}) - \lambda'']] \\ &\quad + \beta\gamma(r - \bar{r})\end{aligned}$$

Solve for the Steady State

- Idea: find time-invariant variables $\bar{w}, \bar{r}, \gamma, K, N, B, A$, policy functions $c(a, e), a'(a, e), \lambda'(a, \lambda, e)$ and distribution μ (denote its pdf as $p(a, \lambda, e)$) that solve the above equations
- Approach
 1. Guess $\bar{w}$
 2. Solve for $\bar{r}(\bar{w})$ which
 - can be used to solve households problems, and
 - generate aggregate variables A, K, B which satisfy the resource constraint
 3. Solve for $\lambda'(a, \lambda, e)$ and γ
 4. Check if the last equation $\partial \bar{w}$ is satisfied, otherwise update on $\bar{w}$

Back

Step 1: Given $\bar{w}$, Find $\bar{r}(\bar{w})$

- Compute N

$$N = \sum_e \pi_e e n(e, \bar{w})$$

- Solve for K

$$1 = \beta (1 + F_K(K, N))$$

- Guess $\bar{r}(\bar{w})$, solve households' problem

$$\begin{aligned} u'(c) &\geq \beta (1 + \bar{r}) \mathbb{E}[u'(c') | e] \text{ with equality if } a' > -\underline{a} \\ c + a' &= a(1 + \bar{r}) + y(e, \bar{w}) \end{aligned}$$

- Use simulation to find the distribution on (a, e) , denote its pdf as $m(a, e)$

Step 1: Given $\bar{w}$, Find $\bar{r}(\bar{w})$

- Compute A

$$A = \sum_e \int a \mu(ds, e)$$

- Compute B

$$B = A - K$$

- Check whether $\bar{r}$ satisfies

$$G + \bar{r}B + \bar{r}K + \bar{w}N \leq F(K, N)$$

Step 2: Solve for $\lambda'(a, \lambda, e) / \gamma$

- Define $q = \frac{\lambda}{\gamma}$ and solve for $q'(a, q, e)$ from

$$\begin{aligned} & u'(c) + u''(c) [q(1 + \bar{r}) - q'] \\ &= \beta(1 + \bar{r}) \mathbb{E} [u'(c') + u''(c') [q'(1 + \bar{r}) - q''] | e] \\ &+ \beta(F_K(K, N) - \bar{r}) \\ &\text{if } a' > -\underline{a}, \text{ otherwise } \lambda' = 0 \end{aligned}$$

- If $a'(a, e) = -\underline{a}$, $q'(a, q, e) = 0$

Step 2: Solve for $\lambda'(a, \lambda, e) / \gamma$

- If $a' > -\underline{a}$, find $q'(\cdot)$ that solves

$$\begin{aligned} & u''(c) [q(1+\bar{r}) - q'(\cdot)] \\ &= \beta(1+\bar{r}) \mathbb{E} [u''(c') [q'(\cdot)(1+\bar{r}) - q'(q'(\cdot))] | e] \\ &+ 1 - \beta(1+\bar{r}) \end{aligned}$$

- Guess $q'(a, q, e) = \alpha_0(a, e)q + \frac{\alpha_1(a, e)}{(1+\bar{r})u''(c)}$ and substitute it into the above equation
- Match the coefficient of q and the constant
- Find $\alpha_0(a, e)$ that satisfies

$$\begin{aligned} & u''(c)(1+\bar{r} - \alpha_0) \\ &= \alpha_0\beta(1+\bar{r}) \mathbb{E} [u''(c')(1+\bar{r} - \alpha'_0) | e] \end{aligned}$$

- Find $\alpha_1(a, e)$ that satisfies

$$\alpha_1 = \alpha_0 [\beta \mathbb{E} (\alpha'_1 | e) - [1 - \beta(1+\bar{r})]]$$

Step 2: Solve for γ

- From

$$\begin{aligned}\partial \bar{r} : A = & \sum_e \int u'(c) q \mu(ds, e) \\ & + \sum_e \int a(u'(c) + u''(c) [q(1 + \bar{r}) - q']) \mu(ds, e)\end{aligned}$$

- Our approach
 - Simulate 20000 periods to compute μ
 - Directly calculate RHS

Details on Solving for γ

- Rewrite A

$$\begin{aligned} A = & \sum_e \int u'(c) \mathbb{E}[q|a, e] m(a, e) da + \frac{1}{\gamma} \sum_e \int a u'(c) m(a, e) da \\ & + (1 + \bar{r}) \sum_e \int a u''(c) \mathbb{E}[q|a, e] m(a, e) da \\ & - \sum_e \int a u''(c) \mathbb{E}[q'|a, e] m(a, e) da \end{aligned}$$

- Solve for $\mathbb{E}[q|a, e] m(a, e)$ by iteration

1. Guess $\mathbb{E}[q|a, e] m(a, e) = f_0(a, e)$
2. Update for $\mathbb{E}[q'|a', e'] m(a', e') = f_1(a', e')$ using

$$\begin{aligned} \mathbb{E}(q'|a, e) m(a, e) &= \alpha_0 \mathbb{E}(q|a, e) m(a, e) + \frac{\alpha_1 m(a, e)}{u''(c)(1 + \bar{r})} \\ \mathbb{E}(q'|a', e') m(a', e') &= \sum_e \pi_{ee'} \frac{\mathbb{E}(q'|a'^{-1}(a', e), e) m(a'^{-1}(a', e), e)}{\frac{d}{da} a'(a'^{-1}(a', e), e)} \end{aligned}$$

until f_i converges.

Step 3: Update $\bar{w}$

- Check whether $\bar{w}$ satisfies

$$\partial \bar{w} : N = (F_N(K, N) - \bar{w}) N'(\bar{w}) \\ + \sum_e \int en(e, \bar{w}) (u'(c) + u''(c) [q(1 + \bar{r}) - q']) \mu(ds, e),$$

again using distribution from simulation

- If not, increase $\bar{w}$ if $LHS < RHS$, and vice versa.
 - One way is to compute the new N and back out a new $\bar{w}$ using

$$N = \sum_e \pi_e en(e, \bar{w}).$$

Back

Details of Steady State Results

	Optimal Steady State (1)	U.S. Calibration (2)
τ_l	49.8%	28.0%
τ_k	21.1%	36.0%
T/Y	21.1%	14.1%
B/Y	1.07	0.62
K/Y	2.89	3.00
$\int n$	0.23	0.33
$N = \int en$	0.31	0.46
$var(\log(y + T))$	0.30	0.51
$var(\log(y + T + \bar{r}a))$	0.35	0.52
coeff. var. a	1.37	1.52
coeff. var. c	0.64	0.80

Understand Steady State Results

- **High labor tax:** 49.8%
 - Insurance
 - Tradeoff against labor supply distortion
- **High government debt:** 1.1 times GDP
 - Relaxes the borrowing constraint
 - Supported by the high labor tax
- Insurance (high labor tax, higher and more equal household asset)
 - Smaller asset variance: coeff. var. (a) $1.52 \searrow 1.37$
 - Smaller disposable income variance $\text{var}(\log y + T + \bar{r}a)$: $0.52 \searrow 0.35$
 - Lower consumption variance: coeff. var. (c) $0.80 \searrow 0.60$

Why High Government Debt?

- Cost: high labor tax
- Benefit: **relax HH borrowing constraint**
 - In addition to the benefit in representative agents model: smooth distortions
- Debt is determined at the margin that the cost equals the benefit
 - High debt: cost is small
 - Unique solution: the additional benefit does not depend on the initial condition
 - Aiyagari and McGrattan (1998)

Robustness - Small Income Effect

- Low risk aversion $\sigma = 1$. Recalibration

	Optimal Steady State (1)	U.S. Calibration (2)
τ_l	47.5%	28%
τ_k	10.6%	36%
T/Y	18.2%	14.1%
B/Y	1.64	0.62
K/Y	3.17	3.00
$\int n$	0.25	0.33
$N = \int en$	0.34	0.46
$var(\log(y + T))$	0.35	0.51
$var(\log(y + T + \bar{r}a))$	0.37	0.51
coeff. var. a	1.27	1.51
coeff. var. c	0.63	0.77

Robustness - Low Labor Supply Elasticity

- Low labor supply elasticity $\phi = 0.5$.

	Optimal Steady State (1)	U.S. Calibration (2)
τ_l	64.6%	28%
τ_k	12.8%	36%
T/Y	21.9%	14.1%
B/Y	2.42	0.62
K/Y	2.71	3.00
$\int n$	0.23	0.33
$N = \int en$	0.30	0.44
$var(\log(y + T))$	0.22	0.51
$var(\log(y + T + \bar{r}a))$	0.31	0.53
coeff. var. a	1.13	1.55
coeff. var. c	0.55	0.75

Robustness: Low Inequality

- Income process from Aiyagari (1994): $\rho = 0.6, \sigma_u = 0.2$

	Optimal Steady State (1)	U.S. Calibration (2)
τ_l	41.7%	28%
τ_k	2.3%	36%
T/Y	14.5%	13.4%
B/Y	1.70	0.62
K/Y	3.37	3.00
$\int n$	0.29	0.33
$N = \int en$	0.31	0.36
$var(\log(y + T))$	0.158	0.163
$var(\log(y + T + \bar{r}a))$	0.160	0.165
coeff. var. a	0.72	0.75
coeff. var. c	0.229	0.230

Robustness: Permanent Productivity Types

- Two types of agents (high v.s. low productivities)

	Optimal Steady State (1)	U.S. Calibration (2)
τ_l	49.2%	28%
τ_k	21.4%	36%
T/Y	20.3%	13.8%
B/Y	1.10	0.62
K/Y	2.90	3.00
$\int n$	0.23	0.33
$N = \int en$	0.32	0.46
$var(\log(y + T))$	0.50	0.61
$var(\log(y + T + \bar{r}a))$	0.51	0.62
coeff. var. a	1.40	1.55
coeff. var. c	0.66	0.81

Algorithm for Solving the Transition

- Similar to how transition dynamics of Aiyagari model is solved
- Guess $\{\bar{w}_t, \bar{r}_t\}_{t=0}^{\infty}$
- Solve the households' problem on $a_{t+1}(a_t, e_t)$ from backward
- Compute $m_t(a_t, e_t), A_t, B_t, K_t$ from forward
- Solve for γ_t and $\lambda_{t+1}(a_t, e_t)$ from backward
- Compute $p_t(a_t, \lambda_t, e_t)$ from forward
- Check and update $\{\bar{w}_t, \bar{r}_t\}_{t=0}^{\infty}$

Back

Step 1: Guess $\{\bar{w}_t, \bar{r}_t\}_{t=0}^{\infty}$ and Solve Households' problem

- Guess $\{\bar{w}_t, \bar{r}_t\}_{t=0}^{\infty}$
- Solve households' problem by backward induction, as in Aiyagari model
- Compute $n(e_t, \bar{w}_t)$ and $y(e_t, \bar{w}_t)$
- Solve for $a_{t+1}(a_t, e_t)$ by backward induction

$$u'(c_t) \geq \beta(1 + \bar{r}_{t+1}) \mathbb{E}[u'(c_t) | e_{t+1}]$$

with equality if $a_{t+1} > -\underline{a}$

where

$$c_t = a_t(1 + \bar{r}_t) + y(e_t, \bar{w}_t) - a_{t+1}$$

Step 2: Compute $m_t(a_t, e_t), A_t, K_t, B_t$

- Compute m_t , starting from $m_0(a_0, e_0)$ using transition matrix

$$\begin{aligned} & m_{t+1}(a_{t+1}, e_{t+1}) \\ &= \sum_{e_t} \pi(e_{t+1}|e_t) \int I[g_{a,t+1}(a_t, e_t) = a_{t+1}] m_t(a_t, e_t) da_t \end{aligned}$$

or simulation

- Compute A_t

$$A_t = \sum_{e_t} \int a_t m_t(a_t, e_t) da_t$$

- Compute K_t and B_{t+1}

$$\begin{aligned} K_t &= A_t - B_t, \\ B_{t+1} &= F(K_t, N_t) - G_t + (1 + \bar{r}_t) B_t + \bar{r}_t K_t + \bar{w}_t N_t \end{aligned}$$

Step 3: Solve for γ_t and $\lambda_{t+1}(a_t, \lambda_t, e_t)$

- Backward induction
- Compute γ_t

$$\gamma_t = \beta (1 + F_K(K_{t+1}, N_{t+1})) \gamma_{t+1}$$

- Solve for $\lambda_{t+1}(a_t, \lambda_t, e_t)$

$$\begin{aligned} & u'(c_t) + u''(c_t)[\lambda_t(1 + \bar{r}_t) - \lambda_{t+1}] \\ &= \beta(1 + \bar{r}_{t+1}) \mathbb{E} [u'(c_{t+1}) + u''(c_{t+1})[\lambda_{t+1}(1 + \bar{r}_{t+1}) - \lambda_{t+2}] | e_{t+1}] \\ &+ \beta \gamma_{t+1} (F_K(K_{t+1}, N_{t+1}) - \bar{r}_{t+1}) \\ &\text{if } a_{t+1} > -\underline{a}, \text{ otherwise } \lambda_{t+1} = 0 \end{aligned}$$

Step 4: Compute $p_t(a_t, \lambda_t, e_t)$

- Forward from $p_0(a_0, \lambda_0, e_0)$ using transition matrix

$$\begin{aligned} & p_{t+1}(a_{t+1}, \lambda_{t+1}, e_{t+1}) \\ &= \sum_{e_t} \pi(e_{t+1} | e_t) \\ & \int \int I[g_{a,t+1}(a_t, e_t) = a_{t+1}, g_{\lambda,t+1}(a_t, \lambda_t, e_t) = \lambda_{t+1}] \\ & p_t(a_t, \lambda_t, e_t) da_t d\lambda_t \end{aligned}$$

or simulation

Step 5: Check and Update $\{\bar{w}_t, \bar{r}_t\}$

- Check whether the last two series of equations are satisfied

$$\begin{aligned}\gamma_t A_t = & \sum_{e_t} \int \int u'(c_t) \lambda_t p_t(a_t, \lambda_t, e_t) da_t d\lambda_t \\ & + \sum_{e_t} \int \int a_t (u'(c_t) + u''(c_t) [\lambda_t (1 + \bar{r}_t) - \lambda_{t+1}]) \\ & p(a_t, \lambda_t, e_t) da_t d\lambda_t\end{aligned}$$

and

$$\begin{aligned}\gamma_t N_t = & \gamma_t (F_N(K_t, N_t) - \bar{w}_t) N'(\bar{w}_t) \\ & + \sum_{e_t} \int \int e_t n(e_t, \bar{w}_t) (u'(c_t) + u''(c_t) [\lambda_t (1 + \bar{r}_t) - \lambda_{t+1}]) \\ & p_t(a_t, \lambda_t, e_t) da_t d\lambda_t\end{aligned}$$

- If not, update $\{\bar{w}_t, \bar{r}_t\}$

Why a Simple Counter-Argument Does Not Work

- Potential counter-argument:
 - Start at the optimal long-run solution.
 - Compute backwards using FOCs of the Ramsey problem.
 - Provides link between initial and terminal condition
- Does not work in incomplete markets models
 - Knowing c_{t+1} and r_{t+1} of a credit-constrained
 $\nRightarrow$ Know c_t .